

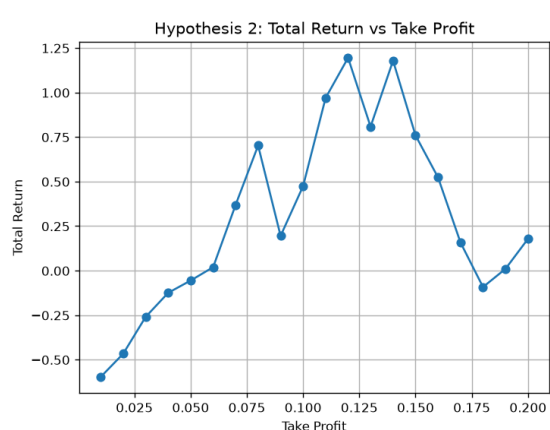
Person A: Momentum/Trend

1. Short-term momentum continuation

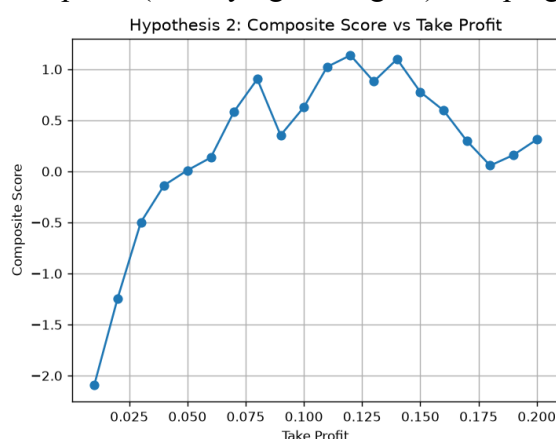
- (1) Hypothesis: the higher the returns in the past 1/2/3 hours, the higher the returns in the next 1/2/3 hours; Signal: past return > threshold
- (2) Data
 - 1 Trading pair: BTCUSDT
 - 2 Train period: 2020-08-01 to 2023-08-31
 - 3 monthly data with 1h candles
- (3) Threshold = 0.1%: Pearson correlation analysis results in slightly negative r , meaning no linear relationship. Bucket analysis confirms that there's no linear relationship.
- (4) When threshold is 0.2%, 0.5%, and 1%, the statistics are about the same.
- (5) Even without doing hypothesis testing, this strategy can create high transaction cost, which is not desirable.

2. Breakout continuation

- (1) Hypothesis: when price is higher than past 24h high, this increasing trend might continue in the short term; Signal: hourly close price > past 24h high * c (c is some constant)
- (2) Data
 - 1 Trading pair: BTCUSDT
 - 2 Train period: 2020-08-02 to 2023-08-31
 - 3 monthly data with 1h candles
- (3) Initially we exit after 1/2/3 hours of entry. When $c = 1$, the number of trades is too large, meaning transaction cost is high and return is negative, so we try to limit the volume by increasing c . It is effective in reducing the volume and transaction cost but return is still negative.
- (4) We tried introducing stop loss and take profit (modifying exit signal). Keeping both



Graph 1



Graph 2

HYPOTHESIS 2 - STOP LOSS / TAKE PROFIT STRATEGY	
Initial capital:	\$100,000.00
Breakout threshold:	0.50%
Stop loss:	-0.50%
Take profit:	+12.00%

Final portfolio value:	\$219,481.55
Total return:	119.48%
Maximum drawdown:	-37.80%
Number of trades:	263
Total fees paid:	\$78,972.90

Sharpe ratio:	1.0031
Sortino ratio:	1.5088
Calmar ratio:	0.7692

Table 1

Detailed Portfolio Statistics =====	
Start	0
End	26979
Period	1124 days 04:00:00
Start Value	100000.0
End Value	219481.546498
Total Return [%]	119.481546
Benchmark Return [%]	120.774296
Max Gross Exposure [%]	100.0
Total Fees Paid	78972.896065
Max Drawdown [%]	37.800643
Max Drawdown Duration	793 days 11:00:00
Total Trades	263
Total Closed Trades	263
Total Open Trades	0
Open Trade PnL	0.0
Win Rate [%]	9.505703
Best Trade [%]	11.788
Worst Trade [%]	-4.370585
Avg Winning Trade [%]	11.788
Avg Losing Trade [%]	-0.835528
Avg Winning Trade Duration	3 days 19:50:24
Avg Losing Trade Duration	0 days 15:12:21.176470
Profit Factor	1.400401
Expectancy	454.302458
Sharpe Ratio	1.003134
Calmar Ratio	0.769203
Omega Ratio	1.074945
Sortino Ratio	1.50881

Table 2

3. Moving-average trend

- (1) Hypothesis: when 6-hour moving average curve crosses 24-hour moving average curve from below, the price will increase in the short term; Signal: MA-6 crosses MA-24 from below

- (2) Data

- 1 Trading pair: BTCUSDT
- 2 Train period: 2020-08-01 to 2023-08-31
- 3 monthly data with 1h candles

- (3) Same as other hypotheses: without risk-managing measures, the returns are very bad.
- (4) AI suggested using dynamic ATR-scaled position sizing, which means define the maximum risk we can tolerate, which determines the maximum position we take (we don't use 100% of the money to trade every time the signal is activated).
- (5) I tested combinations of these values for the four parameters:

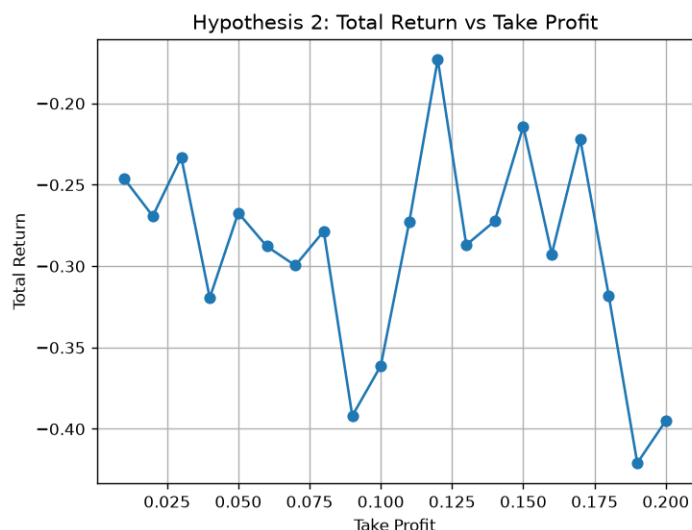
$$\text{Position Value} = \frac{\text{Equity} \times \text{Risk Per Trade}}{\text{ATR} \times \text{ATR Stop Multiplier}} \times \text{Price}$$

ATR_PERIOD	12	24	48
RISK_PER_TRADE	0.5%	1%	2%
ATR_STOP_MULTIPLIER	1.5	2.0	3.0
MAX_POSITION	25%	50%	100%

breakout threshold and stop loss at 0.005, I tested these values for take profit: {0.01, 0.02, 0.03, ... 0.18, 0.19, 0.20} (Graph 1 & 2), the largest total return and composite score appear at approximately 0.12 (Table 1 & 2).

(5) However, this relationship doesn't seem to hold for the test period as shown in graph 3 below (2023-09-01 to 2026-08-01).

(6) The trend that this strategy works for the train period but not the test period holds for the trading pair ETHUSDT.



Graph 3

In general, this parameter combination provides better returns:

- 1 ATR Period = 48
- 2 Risk Per Trade = 0.02
- 3 Max Position = 100%

(6) For “ETHUSDT_1h_2020-08-01_to_2023-08-31.csv”, ATR_STOP_MULTIPLIER = 1.5 creates the greatest return, as shown below (Table 3).

```
Start      2020-08-01 00:00:00+00:00
End        2023-09-01 00:00:00+00:00
Period     1125 days 05:00:00
Start Value      100000.0
End Value       201902.113491
Total Return [%] 101.902113
Benchmark Return [%] 380.456963
Max Gross Exposure [%] 100.0
Total Fees Paid  82523.048071
Max Drawdown [%] 45.809138
Max Drawdown Duration 727 days 07:00:00
Total Trades      22
Total Closed Trades 22
Total Open Trades 0
Open Trade PnL    -513.71177
Win Rate [%]      38.7665
Best Trade [%]    59.61345
Worst Trade [%]   -26.61919
Avg Winning Trade [%] 6.62273
Avg Losing Trade [%] -3.21863
Avg Winning Trade Duration 3 days 18:07:3
Avg Losing Trade Duration 1 days 17:50:56.11510
Profit Factor      1.12704
Expectancy        451.17103
Sharpe Ratio      0.70456
Calmar Ratio      0.55879
Omega Ratio       1.03229
Sortino Ratio     0.99409
dtype: object

===== Parameters & Composite Score =====

Parameters:
Short MA:      48 hours
Long MA:       72 hours
ATR Period:    48 hours
Risk Per Trade: 2.00%
ATR Stop Multiplier: 1.50 x ATR
Max Position:  100.00%
```

Table 3

(7) Good news, this strategy is the first and only strategy that works in both the train and test period (Table 4). Hooray!

(8) It also works for the trading pair ETHUSDT (Table 5).

```
Start      2023-09-01 00:00:00+00:00
End        2026-08-02 00:00:00+00:00
Period     1066 days 01:00:00
Start Value      100000.0
End Value       200692.028473
Total Return [%] 100.692028
Benchmark Return [%] 141.187096
Max Gross Exposure [%] 100.0
Total Fees Paid  92260.090166
Max Drawdown [%] 45.980594
Max Drawdown Duration 436 days 21:00:00
Total Trades      201
Total Closed Trades 201
Total Open Trades 0
Open Trade PnL    0.0
Win Rate [%]      41.293532
Best Trade [%]    29.843048
Worst Trade [%]   -7.037198
Avg Winning Trade [%] 3.654336
Avg Losing Trade [%] -1.854106
Avg Winning Trade Duration 4 days 08:33:58.554216
Avg Losing Trade Duration 1 days 18:20:50.847457
Profit Factor      1.196942
Expectancy        500.955366
Sharpe Ratio      0.95566
Calmar Ratio      0.585799
Omega Ratio       1.043237
Sortino Ratio     1.391677
dtype: object

===== Parameters & Composite Score =====

Parameters:
Short MA:      48 hours
Long MA:       72 hours
ATR Period:    48 hours
Risk Per Trade: 2.00%
ATR Stop Multiplier: 3.00 x ATR
Max Position:  100.00%
Taker Fee:     0.100%
Composite Score: 1.019108
```

Table 4

```
Start      2023-09-01 00:00:00+00:00
End        2026-08-02 00:00:00+00:00
Period     1066 days 01:00:00
Start Value      100000.0
End Value       147691.461235
Total Return [%] 47.691461
Benchmark Return [%] 11.72965
Max Gross Exposure [%] 100.0
Total Fees Paid  55861.167363
Max Drawdown [%] 43.932491
Max Drawdown Duration 298 days 12:00:00
Total Trades      217
Total Closed Trades 217
Total Open Trades 0
Open Trade PnL    0.0
Win Rate [%]      38.709677
Best Trade [%]    30.644771
Worst Trade [%]   -11.144093
Avg Winning Trade [%] 4.666257
Avg Losing Trade [%] -2.593122
Avg Winning Trade Duration 3 days 23:32:08.571428
Avg Losing Trade Duration 1 days 16:40:09.022556
Profit Factor      1.103482
Expectancy        219.776319
Sharpe Ratio      0.568862
Calmar Ratio      0.325134
Omega Ratio       1.025523
Sortino Ratio     0.821663
dtype: object

===== Parameters & Composite Score =====

Parameters:
Short MA:      48 hours
Long MA:       72 hours
ATR Period:    48 hours
Risk Per Trade: 2.00%
ATR Stop Multiplier: 3.00 x ATR
Max Position:  100.00%
Taker Fee:     0.100%
```

Table 5

In conclusion, the third hypothesis with position control is the most promising strategy.

Important lesson:

1. Learning to manage risks through various methods rather than using the raw hypotheses directly is more effective in producing viable strategies that not only work in the train period but could also work in the test period (the so-called “out-of-sample test”).
2. When choosing parameters, it is essential to avoid overfitting, which leads to good performance in the train period but bad performance in the test period. Overfitting could be the result of choosing the exact parameters or combination of parameters that provide peak performance during train period. Therefore, we should identify parameter values that provide consistent good performance (when other parameters change), and whose nearby values also result in good or similar performance.